Fiscal Year:	FY 2016	**Task Last Updated:**	FY 12/08/2016
PI Name:	Burma, Sandeep Ph.D.		
Project Title:	Mouse Glioma Models to Estimate Cancer Risks from HZE Particle Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	78229	**Congressional District:**	21
Comments:	NOTE: Formerly at University of Texas Southwestern Medical Center at Dallas until fall 2019.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	03/11/2013	**End Date:**	09/10/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/10/2016 per NSSC information (Ed., 1/6/16)		
Key Personnel Changes/Previous PI:	January 2016: No key personnel changes.		
COI Name (Institution):	Bachoo, Robert M.D., Ph.D. (University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	NNX13AI13G		
Performance Goal No.:			
Performance Goal Text:			
	There is a paucity of systematic studies carried out with well-defined model systems to evaluate cancer risks to astronauts from exposure to charged particles. Glioblastomas (GBM) are deadly brain tumors with very poor prognosis and are the third leading cause of cancer-related deaths among adults aged 30-50 years. Exposure to ionizing radiation is the only known risk factor for the development of these tumors. Unprecedented advances have recently been made in the genetic analysis of these tumors as part of the Human Cancer Genome Atlas Project. Five key genetic changes predominate in GBMs -- loss of Ink4a, Arf, p53, and/or PTEN, and amplification of EGFR. Under the auspices of our currently-funded NASA grant, our lab has validated sensitive mouse models with brain-targeted deletions of these genes representing progressive steps in gliomagenesis. By using genetically “pre-sensitized” models, we are capable of rapidly assessing brain tumor development after exposure to particle radiation. Our current research shows that Fe ions are		

Task Description:	4-fold more gliomagenic than X-rays. Additionally, we have demonstrated that Fe-induced gliomagenesis is triggered by specific oncogenic events, such as MET amplification or PDGFR over-expression, which are also relevant to human GBMs. We are currently using further finessed versions of these mouse models to analyze not only the tumorigenic potential of an array of particles and doses, but also to identify the cell types in the brain that are most susceptible to transformation following charged particle exposure. Furthermore, we propose to undertake large-scale oncogenomic studies to identify pathways that are altered in particle-induced GBMs. The results of this study will greatly facilitate the estimation of cancer risks from charged particle exposure and should ultimately lead to radioprotective or therapeutic strategies to inhibit or treat these recalcitrant tumors. Specific aims are: Aim 1: Evaluation of the tumorigenic potential of HZE (high energy) particles using Nestin-Cre driven Ink4a/b-/-Arff/f and p53+/f Pten+/f mouse GBM models. Aim 2: Identification of the “cell-of-origin” of HZE-induced gliomas using cell-type specific Cre-ERT2 mouse GBM models. Aim 3: Elucidating mechanism(s) underlying HZE-induced gliomagenesis using an integrated oncogenomic appoach.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As cancer therapy with particles (protons and C ions) is becoming more widely used, our research with heavy ions is also important from a therapeutic standpoint here on Earth.
Task Progress:	SUPPLEMENTAL REPORTING FOR FINAL REPORT DECEMBER 2016: There is a pressing need for systematic studies with well-defined model systems to evaluate cancer risks to astronauts from exposure to charged particles. Glioblastomas (GBM) are deadly brain tumors with very poor prognosis and are the third leading cause of cancer-related deaths among adults aged 30-50 years. Exposure to ionizing radiation is the only known risk factor for the development of these tumors. Unprecedented advances have recently been made in the genetic analysis of these tumors as part of the Human Cancer Genome Atlas Project. Five key genetic changes predominate in GBM -- loss of Ink4a, Arf, p53, and/or PTEN, and amplification of EGFR. As part of this project, we validated sensitive mouse models with brain-targeted deletions of these genes in combinations representing progressive steps in gliomagenesis. We identified two cohorts of mice that do not form spontaneous gliomas but develop high grade malignancies following particle irradiation. The first cohort is deficient in the Ink4a, Ink4b, and Arf genes (Nestin-Cre; Ink4a/b-/-Arff/f) with Nestin-Cre driving brain-specific deletions of conditional alleles. The second model carries Nestin-Cre-driven heterozygous deletions of the p53 and Pten genes (Nestin-Cre; p53+/f Pten+/f). By using these genetically “pre-sensitized” models, we are capable of rapidly assessing brain tumor development after exposure to particle radiation. Following intra-cranial irradiation with 600 MeV/n Fe ions both models developed GBMs with a frequency of approximately 25%. At least four-fold higher doses of X-rays were needed to achieve similar tumor frequencies, indicating an RBE (relative biological effectiveness) of at least 4 for gliomagenesis induced by Fe ions. Multimodal analysis of Fe-induced mouse tumors revealed additional alterations of GBM-relevant tumor suppressor pathways, similar to that seen in humans. Amplification of the receptor tyrosine kinase (RTK) Met was the most frequent oncogenic alteration in the Fe-induced tumors in the Ink4a/b-/-Arff/f model. In the p53+/-Pten+/- model, loss of Pten and p53 tumor suppressors, and up-regulation of the RTK Pdgfr-a and/or Met amplification with concomitant activation of PI3K-Akt signaling was common. Both RTK up-regulation events correlated with high expression of the reprogramming transcription factor Sox2. Ex-vivo studies demonstrated the importance of Sox2 for the maintenance of a cancer stem cell phenotype in the tumor cultures. In additional studies we sought to determine the transforming potential of an array of charged particles (proton, carbon, silicon, iron) with different linear energy transfers (LETs) in order to improve our understanding of the risks from GCR (galactic cosmic radiation) exposure. Here, we used both the Nestin-Cre; Ink4a/b-/-Arff/f and the Nestin-Cre; p53+/f Pten+/f models to better understand the risk of radiation-induced gliomagenesis and to exclude the possibility of model-specific effects. We observed the development of high grade gliomas following irradiation with all tested particles. Mice exposed to lighter ions (0.5 Gy of protons or carbon) formed tumors 5-10% of the time. This frequency increased to 37.5% (Ink4a/b-/-Arff/f) and 50% (p53+/f Pten+/f) following exposure to silicon but then decreased with further increases in Z and LET. These results are suggestive of a linear relationship between LET of charged particles and GBM development, which peaks around an LET of 79 KeV/µm (silicon). The results of this study will greatly facilitate the estimation of cancer risks from charged particle exposure and should ultimately lead to radioprotective or therapeutic strategies to prevent or treat these recalcitrant tumors. ANNUAL REPORTING--JANUARY 2016 We identified two cohorts of mice that do not form spontaneous gliomas but develop high grade malignancies following particle irradiation. The first cohort is deficient in the Ink4a, Ink4b, and Arf genes (Nestin-Cre; Ink4a/b-/-Arff/f) with Nestin-Cre driving brain-specific deletions of conditional alleles. The second model carries Nestin-Cre-driven heterozygous deletions of the p53 and Pten genes (Nestin-Cre; p53+/f Pten+/f). Following intra-cranial irradiation with 600 MeV/n Fe ions both models developed GBMs with a frequency of approximately 25%. At least four-fold higher doses of X-rays were needed to achieve similar tumor frequencies, indicating an RBE (relative biological effectiveness) of at least 4 for gliomagenesis induced by Fe ions. Multimodal analysis of Fe-induced mouse tumors revealed additional alterations of GBM-relevant tumor suppressor pathways, similar to that seen in humans. Amplification of the receptor tyrosine kinase (RTK) Met was the most frequent oncogenic alteration in the Fe-induced tumors in the Ink4a/b-/-Arff/f model. In the p53+/-Pten+/- model, loss of heterozygosity of Pten and up-regulation of the RTK Pdgfr-a with concomitant activation of PI3K-Akt signaling was common. Both RTK up-regulation events correlated with high expression of the reprogramming transcription factor Sox2. Ex-vivo studies demonstrated the importance of Sox2 for the maintenance of a cancer stem cell phenotype in the tumor cultures. In additional studies we sought to determine the transforming potential of an array of charged particles with different linear energy transfer (LET)s (proton, carbon, silicon, iron) in order to improve our understanding of the risks from galactic cosmic radiation (GCR) exposure. Here, we used the Nestin-Cre; Ink4a/b-/-Arff/f mouse and demonstrated gliomagenesis following irradiation with all tested particles. Mice exposed to lighter ions (0.5 Gy of protons or carbon) formed tumors 5-10% of the time. This frequency increased up to 37.5% following exposure to silicon but then decreased with further increases in Z and LET. Ongoing studies with the Nestin-Cre; p53+/f Pten+/f model corroborate the results obtained with the Nestin-Cre Ink4a/b-/-Arff/f

	model. These results are suggestive of a linear relationship between LET of charged particles and GBM development which peaks around an LET of 79 KeV/µm (silicon).
Bibliography Type:	Description: (Last Updated: 12/09/2025)
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Awards	Todorova PK, Hardebeck MC, Alcazar CG, Ilcheva M, Camacho CV, Tomimatsu N, Mukherjee B, Story MD, Bachoo RM, Burma S. "First Place to P. Todorova– NSBRI Dr. David Watson Graduate Student Fellow Poster Award, NASA Human Research Program Investigators' Workshop, Galveston, TX, January 2015." Jan-2015
Awards	Todorova PK, Mukherjee B, Wang X, Guida P, Story MD, McKinnon P, Burma S. "First Place to P. Todorova–NSBRI Dr. David Watson Graduate Student Fellow Poster Award, NASA Human Research Program Investigators' Workshop, Galveston, TX, February 2016." Feb-2016